

Stat 517
Homework # 2

1. Let F have an F distribution with parameters r_1 and r_2 . Argue that $1/F$ has an F distribution with parameters r_2 and r_1 .
2. Let $\bar{X}$ be the mean of a random sample of size n from a distribution that is $N(\mu, 9)$. Find n such that $P(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.90$, approximately
3. Let two independent random samples, each of size 10, from two normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ yield $\bar{x} = 4.8$, $s_1^2 = 8.64$, $\bar{y} = 5.6$, $s_2^2 = 7.88$. Find a 95% confidence interval for $\mu_1 - \mu_2$.
4. Let $X_1, X_2, \dots, X_n, X_{n+1}$ be a random sample of size $n + 1$, $n > 1$, from a distribution that is $N(\mu, \sigma^2)$. Let $\bar{X} = \sum_{i=1}^n X_i/n$ and $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2/(n - 1)$. Find the constant c so that the statistic $c(\bar{X} - X_{n+1})/S$ has the t-distribution. If $n = 8$, determine k such that $P(\bar{X} - kS < X_9 < \bar{X} + kS) = 0.80$. The observed interval $(\bar{x} - ks, \bar{x} + ks)$ is often called an 80% *prediction interval* for X_9 .
5. Let X have a pdf of the form $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, zero elsewhere, where $\theta \in \{\theta : \theta = 1, 2\}$. To test the simple hypothesis $H_0 : \theta = 1$ against the alternative simple hypothesis $H_1 : \theta = 2$, use a random sample X_1, X_2 of size $n = 2$ and define the critical region to be $C = \left\{ (x_1, x_2) : \frac{3}{4} \leq x_1 x_2 \right\}$. Find the power function of the test.
6. Let X have a binomial distribution with the number of trials $n = 10$ and with p either $1/4$ or $1/2$. The simple hypothesis $H_0 : p = 1/2$ is rejected, and the alternative simple hypothesis $H_1 : p = 1/4$ is accepted, if the observed value of X_1 , a random sample of size 1, is less than or equal to 3/ Find the significance level and the power of the test.
7. Let X_1, X_2 be a random sample of size $n = 2$ from the distribution having pdf $f(x; \theta) = (1/\theta) e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. We reject $H_0 : \theta = 2$ and accept $H_1 : \theta = 1$ if the observed values of X_1, X_2 , say x_1, x_2 , are such that

$$\frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} \leq \frac{1}{2}.$$

Here $\Omega = \{\theta : \theta = 1, 2\}$. Find the significance level of the test and the power of the test when H_0 is false.

8. Assume that the weight of cereal in a “10-ounce box” is $N(\mu, \sigma^2)$. To test $H_0 : \mu = 10.1$ against $H_1 : \mu > 10.1$, we take a random sample of size $n = 16$ and observe that $\bar{x} = 10.4$ and $s = 0.4$.
- (a) Do we accept or reject H_0 at the 5% significant level?
 - (b) What is the approximate p-value of this test?